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Does differential predation of *Littorina saxatilis* colour morphs contribute to the maintenance of a colour cline? Insights from a field tethering experiment

Juan Gefaell‡, Ramón Vigo‡ and Emilio Rolán-Alvarez

Centro de Investigación Mariña, Universidade de Vigo, Departamento de Bioquímica, Genética e Inmunología, 36310 Vigo, Spain

Correspondence: J. Gefaell; e-mail: gefaell@uvigo.gal

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ABSTRACT

Colour clines constitute an interesting topic of study for evolutionary ecologists as they allow for the testing of different hypotheses on the adaptive value of colour variation. One possible explanation for the selective advantage of colouration is crypsis. In *Littorina saxatilis* from the Rías Baixas (northwest Iberian Peninsula), a temporally stable and spatially recurrent colour cline has been described. This cline extends from wave-sheltered localities to the most wave-exposed areas, the latter bearing monomorphic populations of a lineated morph. As the ecological communities, and therefore the background colours, change gradually along the wave-exposure continuum, a plausible explanation for this cline is selection for crypsis, by which less cryptic morphs are weeded out from the populations, presumably by crabs. This would be especially intense at the monomorphic localities along the cline. In the present study, we describe a tethering experiment at a representative wave-exposed site from the Rías Baixas (Cabo Silleiro) aimed at testing this hypothesis, in addition to a series of complementary analyses on Cabo Silleiro and a nearby polymorphic site (Aguncheiro), from which most of the snails used in the experiment were collected. The analysis of Aguncheiro revealed slight differences in the distribution of shell scars (reflecting past crab attacks) across colour morphs, without clear support for the selection for crypsis hypothesis. In the tethering experiment, predation rates by crabs were recorded for three different transplanted colour morphs from Aguncheiro, along with the resident morph from Cabo Silleiro. The results were suggestive, but not conclusive. The presumably more conspicuous morph was more likely to be predated in only one of two sessions and only when classifying both shell chips and more substantive breaks as predation attempts. While limited in power, the results do provide valuable data for designing future experiments and motivation for continued investigation of shell colour morph variation in this organism.

INTRODUCTION

Clines, or the gradual progression in a given trait over a geographic transect, have long been an important topic of study for evolutionary ecologists (Huxley, 1938; Endler, 1977; Barton & Hewitt, 1985; Futuyma & Kirkpatrick, 2017). Clines provide insights into how the contrasting forces of genetic drift and spatially varying selection on the one hand, and gene flow on the other, shape the genetic and phenotypic diversity of populations. While genetic drift and, most importantly, spatially varying selection favour the genetic differentiation of populations, gene flow prevents these populations from getting genetically isolated, thus hindering the speciation process and conserving the unity of species (Futuyma & Kirkpatrick, 2017). An interesting type of cline is the group of clines that affects the chromatic properties of the integument of an organism (Antoniazza *et al.*, 2010; McLean & Stuart-Fox, 2014; Köhler, Samietz & Schielzeth, 2017; Gefaell *et al.*, 2024a). Colour clines can result from either a gradual change in the value of a quantitative chromatic variable or a progressive variation in the relative frequencies of a series of discrete morphs (i.e. a morph ratio cline *sensu* Wunderle, 1981). Both kinds of colour clines have been described in most animal groups, spanning from birds and mammals (Sumner, 1926; Mullen & Hoekstra, 2008; Antoniazza *et al.*, 2010; Cosentino, Vanek, & Gibbs, 2023) to insects and molluscs (Saccheri *et al.*, 2008; Brakefield & de Jong, 2011; Gefaell, Galindo & Rolán-Alvarez, 2023).

The colour of an animal can serve at least three different purposes: thermoregulation, communication with conspecifics or predation avoidance (Endler, 1978). This suggests that colour clines might result from gradual changes in the thermal or chromatic profiles of localities along a geographic transect, such that these changes generate differential selective pressures that ultimately drive the evolution and maintenance of the colour cline. Colour clines driven by thermal variation have been described in many

‡Contributed equally.

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species, corroborating the thermal melanism hypothesis (Clusella-Trullas, van Wyk & Spotila, 2007). Likewise, a study on lizards suggests that selection for increased detectability by mating partners can sometimes induce the evolution of a colour cline (McLean, Stuart-Fox & Moussalli, 2015). Finally, differences in the chromatic properties of substrates along a geographic transect have also been shown to favour the establishment of cryptic colour clines in prey species attacked by visual predators (Sumner, 1926; Mullen & Hoekstra, 2008).

Focusing on this last case, colour clines putatively maintained by selection for crypsis can be studied through the use of different methods (see Endler, 1986). Among these, frequentist-observational studies have traditionally been the most used, especially in molluscs (for a review of marine gastropods, see Gefaell *et al.*, 2023). In this approach, the frequencies of colour morphs along a gradient are analysed and compared to the different substrates on which they are found. If the dominant morph at a given site shows a chromatic concordance with its habitat, it is concluded that selection for crypsis is most likely driving the maintenance of the cline.

This frequentist-observational approach has provided suggestive evidence for the role of crypsis in the maintenance of many marine gastropod colour clines. However, its inferential power, that is its capacity to generate robust causal inferences on the mechanisms at work, is limited (Endler, 1986). For this reason, when feasible, either laboratory or field experiments are needed to test whether a particular visual predator acting under a chromatically diverse transect can indeed bring about the evolution of a colour cline in a prey (gastropod) species. Both in molluscs and other taxa, there are some examples of successful laboratory and, especially, field experiments aimed at testing the selective importance of cryptic colouration (Ekendahl, 1998; Webster *et al.*, 2008; Vignieri, Larson & Hoekstra, 2010; de Alcantara Viana *et al.*, 2023). Nonetheless, field experiments, despite their high ecological validity, are particularly difficult to conduct due to the unpredictability and lack of control over extraneous variables inherent in working in the wild. Additionally, these experiments rely on numerous tacit assumptions about the natural history and ecology of the species involved. While evolutionary ecologists devote significant efforts to corroborating some of these assumptions (Lloyd, 1987), many remain untested. This means that if any of these untested assumptions do not hold, the field experiment can fail.

A marine gastropod widely used in evolutionary ecology studies is *Littorina saxatilis* (Olivi, 1872) (Rolán-Alvarez, Boulding & Austin, 2015; Johannesson, 2016). *Littorina saxatilis* inhabits the supralittoral zone of intertidal ecosystems along the northern coasts of the Atlantic Ocean, occupying the ecological niche of a microalgae-grazing feeder (Reid, 1996). Partly due to its ovoviviparous breeding system, *L. saxatilis* has evolved numerous local adaptations (Rolán-Alvarez, 2007; Butlin *et al.*, 2014; Rolán-Alvarez *et al.*, 2015). This mode of reproduction curbs its dispersal capabilities, thereby significantly limiting the erosion of genetic adaptations to specific environments that gene flow can produce (i.e. gene swamping; Futuyma & Kirkpatrick, 2017). In addition, the diversity of intertidal ecosystems in terms of species and substrate composition generates diverging selective pressures to which local populations of *L. saxatilis* adapt (Rolán-Alvarez, 2007).

Among the many traits exhibiting variation in *L. saxatilis*, shell colour stands out (Johannesson & Butlin, 2017; Gefaell *et al.*, 2023, 2024a). This species presents both intrapopulation and geographic colour variation, the latter usually taking the form of morph ratio clines (Sacchi, 1979; Reid, 1996; Gefaell *et al.*, 2024a). An area where these clines are particularly noticeable is the Rías Baixas, a series of four coastal inlets that originated through the immersion of four river valleys following the Holocene transgression (Arce-Chamorro, Vidal-Romaní & Sanjurjo-Sánchez, 2022). The Rías Baixas show a nearly continuous wave-exposed gradient from their inner parts (less wave-exposed) to their outer regions (more wave-exposed) (Gefaell *et al.*, 2024a). This gradient is accompanied by a concomitant gradual progression in the species composition of their intertidal ecosystems (with different algae and sessile and mobile animals), as well as a steady change in the frequencies of the two main *L. saxatilis* colour morphs living there: a fawn-yellowish morph that abounds in the sheltered zones and a lineated morph (*lineata*) whose incidence grows as we approach the most exposed parts, until eventually reaching monomorphism. At intermediate localities, high colour diversity (white, yellow, orange, black, etc.) and less predictable patterns usually occur (Sacchi, 1979; Reid, 1996; Gefaell *et al.*, 2024a).

It has been recently shown that the morph ratio cline of *L. saxatilis* from the Rías Baixas has remained relatively stable over the last 40 years in at least one of these four Rías, the Ría of Vigo, the southernmost and most studied (Gefaell *et al.*, 2024a). The temporal stability of this cline, in addition to its spatial replication in the four Rías (Reid, 1996), suggests that it is being maintained by spatially varying selection (Endler, 1986). The specific selective pressures at work, though, are not yet known. Of the three functions colour might have, mate recognition and conspicuousness are highly unlikely due to the lack of colour vision in *L. saxatilis* (see Gefaell *et al.*, 2024b, for an experimental exploration of sexual selection in *L. saxatilis* colour morphs). Thus, thermal stress and selection for crypsis may contribute to clinal variation in this organism. Here, we focus on selection for crypsis, which has been described in many marine gastropods (Gefaell *et al.*, 2023, and references therein).

Informal observations of *L. saxatilis* colour morphs in their natural environments suggest crypsis, as the dominant morphs across localities in the Rías Baixas region tend to blend chromatically with their backgrounds (J. Gefaell, personal observation). Furthermore, most of *L. saxatilis*' predators (Pettitt, 1975), particularly shorebirds, crabs and intertidal percomorph fishes (such as blennies and gobiids), rely on vision to varying extents when hunting for prey. If the chromatic properties of localities along the Rías Baixas are heterogeneous due to the different ecological communities they contain, this can favour the evolution of a cryptic colour cline in a prey species such as *L. saxatilis*. This selective pressure may be especially intense in areas exhibiting monomorphic populations, such as those of the *lineata* morph in wave-exposed localities, where it can presumably weed out all individuals not displaying the most cryptic colouration. The *lineata* morph, in contrast, might be highly cryptic against the barnacle-covered and bare alkaline granitoid rocks characteristic of these localities, thus more efficiently avoiding visual predators.

To test this hypothesis, we conducted a tethering experiment in a representative *lineata* monomorphic wave-exposed site from the Ría of Vigo (Cabo Silleiro) using three different transplanted (i.e. allochthonous) *L. saxatilis* morphs of varying degrees of presumed crypsis (yellow, black and *lineata*) from a nearby polymorphic population (Aguncheiro). To these, *lineata* individuals from the study site (resident i.e. autochthonous, and presumably highly cryptic) were added to account for transplantation effects. Snails from all these morphs were tethered at points of seemingly high marbled rock crab (*Pachygrapsus marmoratus*) density to record successful or failed predation attempts by this species, as revealed by the resulting marks on the snails' shells. If selection for crypsis is at work, we would expect increased predation of the less cryptic morphs (yellow and black, especially the former) compared to the *lineata* snails (particularly the resident ones). This would help explain why wave-exposed populations of *L. saxatilis* tend to be monomorphic for *lineata*, thus providing a partial account of the colour cline observed in the Rías Baixas. As a complementary study on the role of selection for crypsis, the frequencies of scars (reflecting past unsuccessful crab attacks) on a sample of the morphs from the source site (Aguncheiro) were counted and a series of key shell traits were measured. Particularly, the degree of crypsis of the different morphs used in the experiment was assessed using reflectance spectrometry.

Reflectance spectrometry corroborated that *lineata* snails (particularly residents) are more cryptic than the black and yellow morphs

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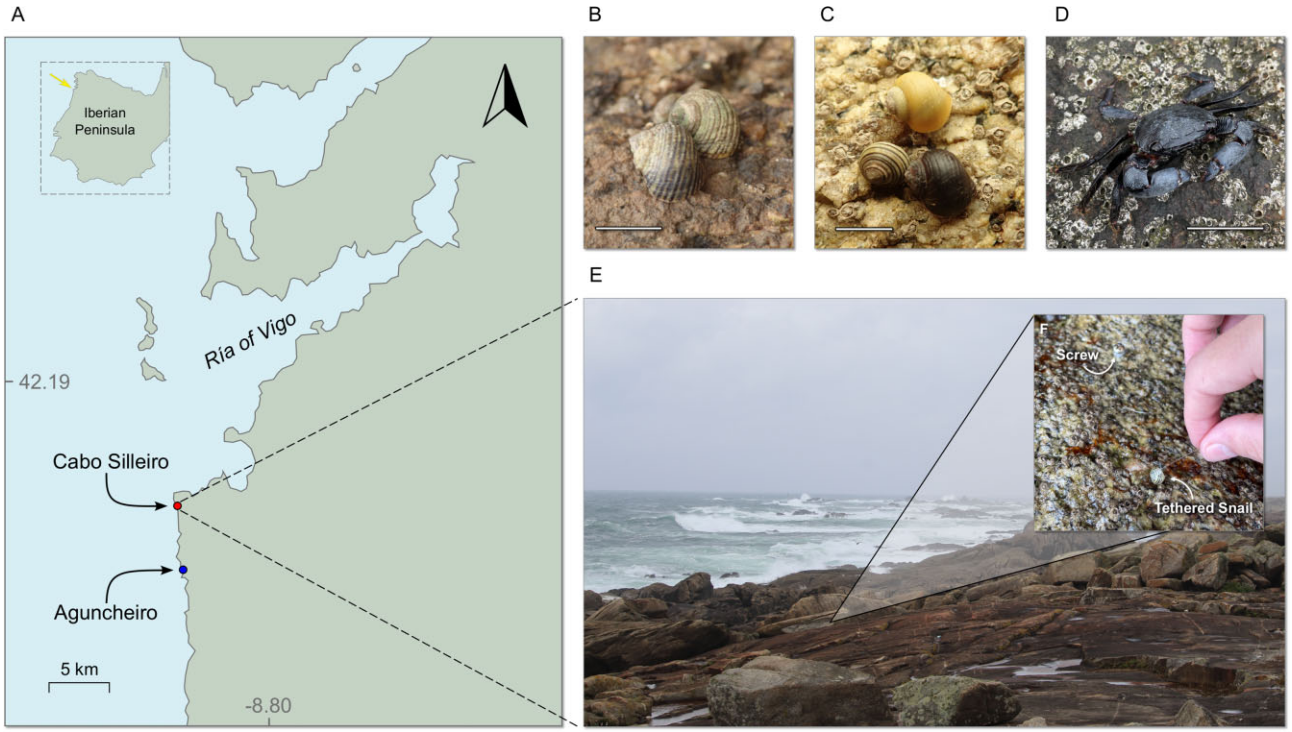


Figure 1. Key aspects of the tethering experiment. **A.** Sites in the Ría of Vigo. **B.** Autochthonous *lineata* morph (from Cabo Silleiro). **C.** Allochthonous *lineata*, *nigra* and *lutea* morphs (from Aguncheiro). **E.** The main predator species studied, the crab *Pachygrapsus marmoratus*. **E**. The site at Cabo Silleiro where the experiment was carried out with the inset **F** showing an allochthonous *lineata* (from Aguncheiro) snail tethered to a screw. Scale bars: **B**, **C**, 7.0 mm; **D**, 4.5 cm.

at the experimental site (Cabo Silleiro). However, the pattern of shell scars on the different morphs at their source site (Aguncheiro) did not yield clear support for the selection for crypsis hypothesis. In the first session of the tethering experiment, there was significant evidence of differential predation; in the second session of the experiment, in contrast, there was no significant evidence, but power was reduced due to low rates of predation overall. These results provide some limited support for selection for crypsis as a contributing mechanism to the maintenance of the *L. saxatilis* colour cline in the Rías Baixas. Nevertheless, they also highlight the challenges of using field approaches to study predation and adaptation, as well as the relevance of statistical power.

MATERIAL AND METHODS

Localities, predator species and morphs studied

The tethering experiment was carried out in the Cabo Silleiro site (Oia, Galicia, Spain; 42.10467, −8.89885; see Fig. 1A). This constitutes a 200-m-long and 80-m-wide dark brown alkaline granitoid bedrock with small rocks on it that experiences intense swell throughout the year (Fig. 1E). The lower parts of the bedrock surface are fully covered with barnacles (*Semibalanus balanoides* and *Chthamalus* spp.), with this barnacle coverage gradually receding as we approach its upper part. In terms of colour, and excluding the low shore *Wave* ecotype, the population of *Littorina saxatilis* inhabiting Cabo Silleiro is monomorphic for *lineata*, with snails showing a white to creamish background with superimposing black lines or stripes that run perpendicular to the lines of shell growth (Fig. 1B). This morph, which in wave-exposed localities can be equated to the *crab* ecotype in the *L. saxatilis* ecotype literature, is presumably cryptic in dark alkaline granitoid rocks (with or without high barnacle coverages), but this hypothesis has not been previously tested. The *L. saxatilis* population from Cabo Silleiro is relatively dense, and previous investigations (e.g. Boulding *et al.*, 2017) suggest that it is subjected to intense crab predation.

One main agent responsible for such predation is the marble rock crab (*Pachygrapsus marmoratus*; Fig. 1D), present at high densities in Cabo Silleiro, where they are especially abundant in the upper part of the ecosystem. These crabs constitute one of the main selective pressures driving the phenotypic and genetic differentiation of the crab and wave ecotypes of *L. saxatilis* in the Galician coasts and have thereby been the subject of numerous investigations over the last few decades (e.g. Rolán-Alvarez, 2007). However, their potential role in maintaining the *lineata* colour monomorphism seen in the wave-exposed localities of the Rias Baixas through the eradication of less cryptic colour morphs of *L. saxatilis* has not yet been assessed.

A crucial question is whether *P. marmoratus* can see colours, something for which there is no direct evidence. Nevertheless, neurophysiological and behavioural studies on other brachyuran species suggest that this group of animals has a chromatic vision. For example, both the blue crab (*Callinectes sapidus*) and the green shore crab (*Carcinus maenas*) have two photoreceptors, one with a peak sensitivity at 508 nm (green), and another with a peak sensitivity at 440 nm (blue; only in the ventral portion of the eye) (Martin & Mote, 1982; Baldwin & Johnsen, 2012). This implies that these crabs can detect chromatic signals through the opponency of the green and blue photoreceptors. This has been confirmed in behavioural experiments, which show that at least *C. sapidus*, as well as other crab species, can use visual information to carry out decisions (Detto, 2007; Baldwin & Johnsen, 2009, 2012; Kawamura *et al.*, 2020). Furthermore, molecular evidence in the sand fiddler crab (*Uca pugilator*) suggests that this species could even be trichromat, as its members have three opsin-encoding genes that appear to have different wavelength sensitivities (Rajkumar *et al.*, 2010). Taken together, this information makes it likely that *P. marmoratus* can see colours. But even if they do not or if this ability is of little use when hunting because of their tendency to be more active at night (as has been shown in Portuguese populations of this species; Cannicci, Paula & Vannini,

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1999), *P. marmoratus* could still respond to achromatic cues based on the signal intensity of its most abundant (or only) photoreceptor. Extrapolated to the present experiment, a plausible assumption is, therefore, that *P. marmoratus* can distinguish between snail colour morphs based on either chromatic (via opponency) or achromatic (intensity) signals, or both.

For the experiment, different colour morphs of *L. saxatilis* present at a nearby site, Aguncheiro (Oia, Galicia, Spain; 42.050963, −8.885492; Fig. 1A), were used. This site, right at the mouth of the Mougás River (*c.* 6 km from Cabo Silleiro), is composed of small to medium (*c.* 1 m in diameter) yellowish alkaline granitoid rocks with black-reddish patches showing a full barnacle coverage in its lower parts that decreases to *c.* 30–50% in the upper areas (see Supplementary Material Fig. S1A). Despite being surrounded by highly wave-exposed areas, the population of *L. saxatilis* from this site is made up of four colour morphs: white or creamish with superimposed lines (*lineata*), black or very dark brown (*nigra*), yellow (*lutea*) and white (*albida*) (see Fig. 1C for the first three morphs). All colour morph names within parentheses follow Sacchi (1979) and are used hereafter. Of the four colour morphs present at Aguncheiro, three of them were used in the experiment, *lineata*, *nigra* and *lutea*. *Albida* was discarded because of its low frequency. To these, a subsample of *lineata* from Cabo Silleiro (autochthonous in the site in which the experiment was carried out) was also used to account for the effects of transplantation and autochthony of the snails.

Tethering experiment design and procedure

The differential predation of snail morphs by crabs was tested with a tethering experiment. The tethering paradigm consists of attaching snails to nylon threads using a water-resistant instant epoxy glue, which are in turn tethered to screws that have been previously installed at spots with high crab predation (one snail per screw). After an (ideally pre-established) amount of time, the snails tethered to the screws are recaptured and the shell marks indicating successful or unsuccessful crab predation attempts are counted. Despite their shortcomings (such as potential inflation of mortality rates in predator-outrunning prey; Barbeau & Scheibling, 1994; Peterson & Black, 1994; Zimmer-Faust *et al.*, 1994; Baker & Waltham, 2020), tethering experiments are one of the few experimental designs that allow for measuring predation rates of littorinids and other marine organisms in the wild (Aronson & Heck, 1995; Yamada & Boulding, 1996; Boulding, Holst & Pilon, 1999; Boulding *et al.*, 2017).

A total of 284 snails, 71 per morph (*lineata* autochthonous and allochthonous, *nigra* and *lutea*), were collected in two batches for the two sessions that comprised the experiment. The samplings were carried out on two non-consecutive days in September and October 2022, on the days before each experimental session. The size of the snails was standardized as much as possible during the sampling sessions by collecting only intermediate-sized snails with the overall range in shell length (measured from apex to the tip of the last whorl) being <0.5 mm across all the snails collected (i.e. shell length varied from 7 to 7.5 mm). Smaller snails (like many utilized by Boulding *et al.*, 2017) were discarded as the glue employed to tether them, despite being transparent, often covered most of the shell, potentially masking their original colour. The integrity of the shell of the snails collected was also ensured. Once picked up, the snails were taken for processing to the facilities of the University of Vigo, Galicia, Spain (25–30 km from the sites). Transportation was done in plastic boxes with rocks from their respective locations to ensure the well-being of the snails. All the specimens were handled and prepared alive for the experiment (see Supplementary Material Fig. S1B) and stored for 16 h in a refrigerator until the experimental sessions began.

A total of 11 transects of either 12 or 28 screws (with at least 1 m between screws), each situated at different spots of the Cabo Silleiro site, were used in the tethering experiment (for exact location of transects see: https://www.google.com/maps/d/edit?mid=1CFlEfOUPsXpst7W7tYs3xFD2RDuAEbA&usp=sharing). Of these, eight transects were specifically set for this experiment (see Supplementary Material Fig. S1C), while the three remaining ones were re-used from the tethering experiment described in Boulding *et al.* (2017) (Supplementary Material Fig. S1D). All transects were set up near crevices where *P. marmoratus* had been previously spotted, along the continuum from the mid-lower to mid-upper parts of the ecosystem (corresponding to what has been reported for *P. marmoratus* from southern Portugal, where this crab is present mostly at the eulittoral and littoral fringes; Cannicci *et al.*, 1999).

The snails used in the experiment were tethered alive to the screws of each transect (Fig. 1F and Supplementary Material Fig. S1E). Most of the resident snails present in the area around the screws were moved outside the transects to increase the chance of the tethered snails being preyed upon. The distribution of the colour morphs was semi-random, with the autochthonous and allochthonous *lineata* individuals separated from each other by either a *nigra* or *lutea* snail to avoid sensory biases in the crabs but allowing the occurrence of *nigra* and *lutea* to vary randomly. The length of the nylon threads used was 30 cm, so the inflated mortality bias of tethering experiments (according to which mortality rates are inflated because the prey cannot outrun or outmanoeuvre the predator) was minimized. However, littorinid snails are apparently not affected by this bias, as no effect was found of tether length (3 *vs* 30 cm) on predation rates in two species of the genus *Littorina* (Rochette & Dill, 2000). Immediately after tethering each snail, the shell length was measured to the nearest 0.1 mm using a Vernier caliper to add a statistical control to the procedural standardization of this variable (Supplementary Material Fig. S1F).

The experiment was divided into two sessions conducted between September and October 2022. The season of the experiment was chosen following Boulding *et al.* (2017), who obtained a relatively high level of predation of *L. saxatilis* by *P. marmoratus* in a tethering experiment carried out during autumn at the same site. Based on a pilot experiment in which the three Aguncheiro morphs (allochthonous *lineata*, *nigra* and *lutea*) were used, it was predetermined that the experiment would last 4 days; due to bad weather and lack of predation during the first days, however, the first session was extended 1 more week.

Once each of the experimental sessions was over, the outcome for the tethered snails was ranked *in situ* using a series of pre-established operational categories based on Boulding *et al.* (2017) (see Supplementary Material Table S1). The categories ranking the various outcomes covered the full range of possibilities for the snails. Of these, six categories were the most relevant: 'alive intact' (alive and without any predation marks), 'preyed alive' (alive with predation marks), 'preyed dead' (dead with predation marks), 'chipped' (the outer lip of the shell showing small or smooth, yet clearly noticeable, breaks but of a magnitude that cannot be readily attributed to a predation attempt by a normal-sized crab; i.e. one 3–4 cm in carapace length; as in Reimchen, 1982: fig. 2A), 'empty shell' (shell intact without snail, possibly due to predation by an animal that can remove the snail without breaking the shell, like a percomorph fish; Reimchen, 1982) and 'epoxy only' (only the tethering line with the epoxy glue found i.e. snail detached) (all categories except empty shell figured in Supplementary Material Fig. S2A–E).

Once the outcome of the snails was assessed, all the living specimens (including 'preyed alive') were untethered from the screws, their shells cleaned of residual epoxy glue and carried alive in plastic boxes with rocks to their original localities (Aguncheiro in the case of allochthonous *lineata*, *nigra* and *lutea*, and the exact spots at Cabo Silleiro where they were originally picked up in the case of autochthonous *lineata*). All the tethering lines were also removed from the screws.

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Frequency of scars and shell measurements of the studied colour morphs

Sampling of the four morphs inhabiting Aguncheiro (*lineata*, *nigra*, *lutea* and *albida*), as well as of *lineata* from Cabo Silleiro, was also carried out for the present study (March 2024). The purpose was to independently explore the predation intensity at each site, as well as the susceptibility of the different morphs to predation. This sampling involved counting the number of individuals of each morph with at least one shell scar, at each site. Shell scars, defined as deformations of the line of growth below which new shell material grows, can provide clues to the local predator abundance (Butlin *et al.*, 2014; Stafford, Tyler & Leighton, 2015). Additionally, if systematic biases exist in the frequency of scars between a series of coexisting colour morphs, then these scars can provide information on the differential predation rates, and therefore the fitness, of different morphs. The sampling procedure involved selecting a series of random patches in the intertidal zone where *L. saxatilis* settles, using a 60-cm-diameter sampling circle to demarcate each area, and collecting all the snails within the circle (i.e. a stratified random sampling procedure). Once picked up, the snails were colour-classified *in situ* and possible shell scars on any of the shell whorls were inspected. After the sampling was finished, the snails were immediately returned to their initial spots so as to minimize compromising the population's viability.

Additionally, several snails representative of each of the colour morph used in the experiment (autochthonous and allochthonous *lineata*, *nigra* and *lutea*; $n = 40$, 10 snails per morph) were collected from Cabo Silleiro and Aguncheiro and taken to the laboratory to measure their shell thickness, shape and reflectance spectrometry. These measurements aimed to test possible differences in shell thickness and shape between colour morphs, which might account for potential differences in predation propensity, as well as to formally assess their degree of crypsis. The specimens picked up were size-standardized as much as possible by choosing individuals between 7 and 7.5 mm in length to replicate the sampling process of the tethering experiment. The collected snails were euthanized using cold temperature (i.e. cold anaesthesia) by putting them into the freezer upon arrival at the University of Vigo facilities. Shell shape was analysed using two-dimensional geometric morphometrics based on photographs of each individual (see Supplementary Material for a detailed description of the geometric morphometrics protocol followed). Shell thickness was measured at the lip of the shell (analogous position in *L. saxatilis* to the one measured in the girdled dogwhelk in Martin, Clusella-Trullas & Robinson, 2022) to the nearest 0.01 mm using an electronic gauge. Previous evidence suggests that there are no differences in shell thickness between the colour morphs of a given population of *L. saxatilis* (Raffaelli, 1979). However, as in the present experiment samples from two populations were used (Cabo Silleiro: autochthonous *lineata*; Aguncheiro: allochthonous *lineata*, *nigra* and *lutea*), measurements of this variable were collected to formally evaluate such a possibility.

Lastly, the reflectance spectra of these very same specimens, as well as of random points of two representative rocks from each site (Cabo Silleiro and Aguncheiro), were recorded to assess the degree of crypsis shown by each of the morphs in their autochthonous and allochthonous habitats. These measurements were primarily used to verify a tacit assumption of our experiment, namely, that in the wave-exposed localities of the Rias Baixas (such as Cabo Silleiro, where the experiment was carried out), the degree of crypsis of the different colour morphs of *L. saxatilis* is *lutea* < *nigra* < *lineata*, so that crabs would probably detect and prey on *lutea* more frequently than on *nigra* and *lineata* (both allochthonous and autochthonous). Additionally, the crypsis of the Aguncheiro morphs against their original background was assessed to aid in the interpretation of shell scars at this site. The reflectance spectra of the shells and the rocks were obtained with an OceanOptics DH-mini, UV-VIS-NIR Lightsource reflectance spectrometer (Ocean Optics Inc., Dunedin, FL, USA), calibrated with a white diffuse reflectance standard and measured at a standardized distance of 3 mm under constant lighting conditions. The relative angle of the measured objects to the probe was 90°. The snails were measured in the central area of the main whorl, while the rocks were measured at three randomly selected surface spots. This was done because classic definitions of crypsis involve concealment against a random sample of the background substrate (Endler, 1978). Integration time was 2,800 ms, and the spectra were averaged to 6 scans, with a boxcar width of 10. The resulting reflectance spectra were processed with the R package 'pavo' v. 2.9.0 (Maia *et al.*, 2013). The spectra acquired were smoothed fitting in a locally estimated scatterplot smoothing (LOESS) regression (span = 0.7) and negative values were corrected by converting them to 0.

Statistical analyses

All the statistical analyses were carried out in R version 4.3.3 (R Core Team, 2023). The frequencies of shell scars in the colour morphs from the source site (Aguncheiro) were studied using independence chi-square and Fisher's exact tests (Sokal & Rohlf, 1995). Chi-square was preferentially used; if the contingency table contained cells with relatively low expected values (*c.* 5), which can produce unreliable results, Yates' continuity correction was applied by default to the chi-square test to reduce the chance of a false positive (but see Sokal & Rohlf, 1995). For 2 × 2 or larger tables with very low expected values (well below 5), Fisher's exact test was used. Fisher's exact test is typically used for 2 × 2 contingency tables with small sample sizes but can be applied to larger tables as well, with the *P*-value being estimated with Monte Carlo simulations to speed up the computation process. Monte Carlo simulations generate data under the null hypothesis of independence and compare the observed test statistic to the distribution of test statistics generated via the simulations (Poldrack, 2023).

The predictive power of several potential explanatory variables on the main outcomes of the experiment was explored by fitting generalized linear models with binomial error structure and logit link function (i.e. logistic regressions; Sokal & Rohlf, 1995). For these models, dichotomous variables were created for each of the main outcomes (e.g. variable 'chipped': 1 = chipped, 0 = not chipped). The 'preyed' outcome variable was obtained by grouping the 'preyed alive' and 'preyed dead' outcomes, as both represent instances of crab predation irrespective of whether the snail survived the attack. An additive category of 'preyed + chipped' was created by coding the 'preyed' and 'chipped' outcomes as '1' and the remaining outcomes as '0'. This variable assumes the hypothetical scenario that 'chipped' represents a predation attempt (i.e. probably by a small crab). For the different models, cases whose original outcome was 'epoxy only' or 'lost' were considered NAs and excluded from the analyses, since we do not know whether these snails were preyed upon or not, due to their loss (thus avoiding the 'missing as predation' bias in tethering experiments; Baker & Waltham, 2020). As for the predictors, these included the colour and source site of the snails (combined into a single variable so as not to overfit the models), shell length, position of the transect along the intertidal ecosystem ('lower' *vs* 'upper' transects, based on the distance to the ocean and barnacle coverage), and finally, the experimental session, as the weather was different between the two sessions and this could have influenced the predation by *P. marmoratus*. The colour morph variable was converted to a factor, with *lineata* from Cabo Silleiro (autochthonous) used as a reference level to assess the impact of the remaining colour morphs in relation to the different outcomes. All the models were fitted using maximum likelihood estimation and convergence was studied using the default Fisher scoring iterations. The fit of the different models was evaluated with the Akaike information criterion (AIC). The significance of the potential

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Table 1. Frequencies of morphs and scars, as well as their corresponding percentages, in Aguncheiro (the site where most of the snails for the tethering experiment were collected) and Cabo Silleiro (where the experiment was conducted).

Site	Colour morph	Morph counts	Scar counts	Relative incidence of scars
Aguncheiro	*Lineata*	44 (49.44%)	9 (20.45%)	40.9%
	Nigra	32 (35.96%)	6 (18.75%)	27.27%
	Lutea	11 (12.36%)	5 (45.45%)	22.73%
	Albida	2 (2.25%)	2 (100%)	9.09%
	Sample size (*n*)	89	22 (24.72%)	–
Cabo Silleiro	*Lineata*	123 (100%)	34 (27.64%)	–

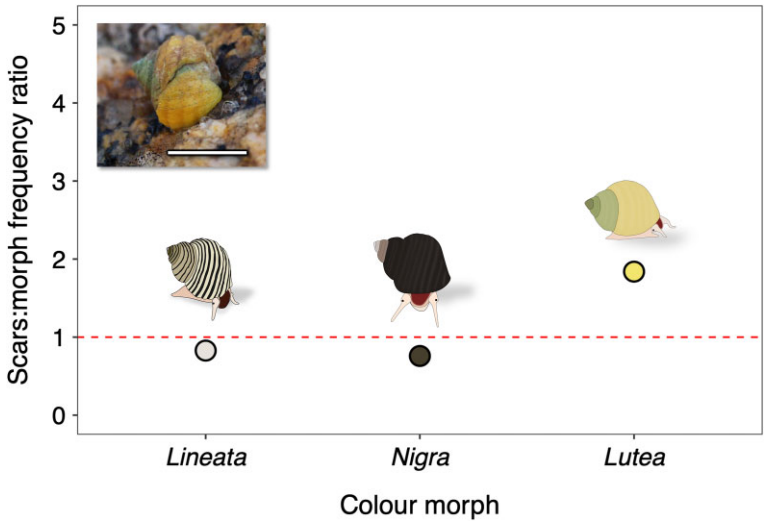


Figure 2. Scars to morph frequency ratio for each colour morph of Aguncheiro (excluding *albida*, which was not used in the tethering experiment and had a low sample size in the samplings). Values well above 1 indicate a higher incidence of scars than the one expected by the frequency of the morph in the population alone. The difference in scars between these morphs, however, is not statistically significant. The inset picture shows a *lutea* snail with a shell scar. Scale bar: 7.0 mm.

predictors was studied through $\tilde{\chi}$- and *P*-values (significance threshold: $P < 0.05$). Statistical power in the logistic regression models was assessed by calculating the odds ratio for each predictor or level, as this measure is commonly used to evaluate effect size for binary variables (Poldrack, 2023). All the models were fitted using the 'glm' function from the R package 'stats' v. 4.5.0. The results from the logistic regressions were further explored through various chi-square and Fisher's exact tests following the same logic described earlier.

Ethical standards

Since *L. saxatilis* is not a protected species and shows relatively dense populations along the Rias Baixas, no special permissions were required for the present study. Nevertheless, we acted in compliance with the ASAB guidelines when handling the snails (ASAB Ethical Committee/ABS Animal Care Committee, 2023). The experiment was deliberately planned to return all the used specimens to their original localities or spots after the sessions, and due to the relatively low predation rate we obtained, most of the snails survived the experiment. Sampling of Aguncheiro and Cabo Silleiro aimed at determining scar frequencies was explicitly designed so as not to compromise the viability of the populations. Finally, the sample of snails collected for the shell measurements, which entailed the death of these individuals, was the lowest possible to ensure a minimum of statistical power.

RESULTS

Frequencies of scars at the source and experiment localities (Aguncheiro and Cabo Silleiro)

A total of 89 (Aguncheiro) and 123 (Cabo Silleiro) snails were collected in six random samplings. Cabo Silleiro was, as expected, monomorphic for *lineata*, while Aguncheiro showed the following frequencies: *lineata* (49.44%) > *nigra* (35.96%) > *lutea* (12.36%) > *albida* (2.25%) (see Table 1). In Aguncheiro, the frequency counts of each morph (*albida*, *lutea*, *nigra* and *lineata*) were homogeneous between the six samplings ($P = 0.493$, Fisher's exact test with *P*-value based on 2,000 Monte Carlo simulations; henceforth used always in Fisher's exact tests), so can be grouped for further analyses. As for predation intensity, indirectly assessed through the frequency of snails with shell scars, both localities have very similar values (Aguncheiro = 24.7%, Cabo Silleiro = 27.6%; χ^2 on count data = 0.10, df = 1, $P = 0.750$).

In Aguncheiro, the percentages of shell scars follow an almost exact reverse pattern than the frequencies of colour morphs: *albida* (100%) > *lutea* (45.45%) > *lineata* (20.45%) > *nigra* (18.75%). Figure 2 shows the ratio of the relative incidence of scars on a morph (i.e. of the total snails with scars, how many of them belong to a given morph) to its relative frequency in the population, excluding *albida*, whose sample size ($n = 2$) is too low to draw robust inferences. Values *c.* 1 indicate that the number of scars in a given morph is the one

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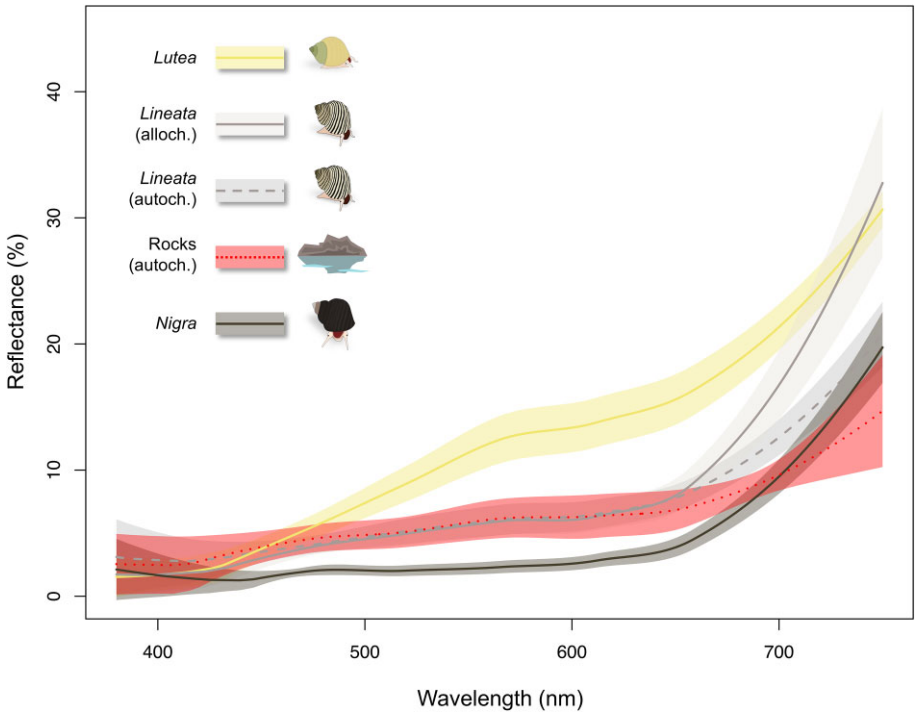


Figure 3. Spectra of objects studied in the experiment. Comparison of the spectra of a sample of snails from the same colour morphs as those studied in the experiment (autochthonous and allochthonous *lineata*, *nigra* and *lutea*) and a random sample of rocks from Cabo Silleiro, 'Rocks (autoch.)'. Each line represents the mean of all measured objects within a class, and the shaded area their corresponding 95% confidence intervals.

that would be expected by its frequency. We see this in both *lineata* and *nigra*, with values slightly lower than 1. Conversely, values significantly over 1 imply a higher rate of scars than that expected by its frequency, as seen in *lutea*. The difference in the frequency of scar counts between colour morphs was statistically significant (Fisher's exact test: $P = 0.031$). However, if we exclude *albida*, which occurred at a very low frequency, the test is not significant (Fisher's exact test: $P = 0.216$).

Shell measurements of the colour morphs

No differences between the colour morphs used in the experiment (autochthonous and allochthonous *lineata*, *nigra* and *lutea*) were found in shell shape, assessed using relative warp 1 (ANOVA: $F_3 = 0.76$, $P = 0.524$) (see Supplementary Material Figs S3 and S4). Conversely, a Kruskal–Wallis test for shell thickness between colour morphs yielded near statistically significant values ($H(3) = 7.31$, $P = 0.062$), with *lutea* showing slightly thicker shells (see Supplementary Material Fig. S5). Effect size, measured through a non-parametric eta-squared for Kruskal–Wallis (η^2_H), was calculated to assess whether the previous result could be due to the low sample size used ($n = 40$), which can lead to a false-negative result (Bateson & Martin, 2021). This test yielded a moderate effect ($\eta^2_H = 0.12$), suggesting that there might be real differences in shell thickness between the colour morphs.

The comparison between the spectra of the rocks and snails from Aguncheiro suggests that, at this site, *lutea* and *lineata* might be more cryptic than *nigra*, as inferred from the overall shape of the spectra lines and their relative reflectance intensities (see Supplementary Material Fig. S6). At the site of the experiment (Cabo Silleiro), however, *lutea* is much more conspicuous than *nigra* and *lineata* (both allochthonous and autochthonous morphs but especially the latter) (see Fig. 3). The *lineata* morphs (both allochthonous and autochthonous) are the ones that overlap the most with the rocks (from *c.* 380 up to 650 nm, the point at which the spectra start to diverge, with autochthonous *lineata* diverging less dramatically). The *lutea* morph stands above the spectrum of the rocks at most wavelengths (higher intensity, thus being brighter), while that of *nigra* stands below it (i.e. less intensity, darker). The distance from these spectra to that of the rocks, however, is apparently less dramatic in *nigra* than in *lutea*.

Tethering experiment

Table 2 summarizes the characteristics and outcomes of the snails used in the two sessions and the whole experiment (sessions 1 and 2 pooled). In the whole experiment, all but 27 snails were recovered and ranked, with the remaining being lost due to unknown reasons or bearing a highly ambiguous outcome (original NAs = 4). The most frequent outcome in the experiment was 'alive', which groups all snails that completed the experiment alive and with their shells undamaged ('alive intact' + 'recovered alive intact' pooled). The frequencies of the remaining outcomes were comparatively low. As for the 'preyed' outcome, only 9% of snails in session 1 ([10/111] × 100) and 6.61% in session 2 ([8/121] × 100; in both cases excluding 'lost (all)' cases, which group 'epoxy only', 'lost', and NAs) were attacked by crabs (results pooled = 7%).

The outcomes were different in each of the experimental sessions (Fisher's exact test: $P = 0.029$; see also Fig. 4). Pairwise comparisons revealed that irrespective of the morph, there were more 'chipped' ($\chi^2 = 6.49$, df = 1, $P = 0.011$) snails in session 1 than in session 2, but no differences in 'preyed' ($\chi^2 = 0.73$, df = 1, $P = 0.395$) or 'empty shell' (Fisher's exact test: $P = 1$). As for the summed results of 'preyed' and 'chipped' (i.e. 'preyed + chipped'), a higher count was also observed in session 1 than in session 2 ($\chi^2 = 7.28$, df = 1, $P = 0.007$). In addition to outcomes, the shell length of the snails involved in the experiment varied between sessions (session 1 = 7.41 ± 0.62 mm, session 2 = 7.19 ± 0.64; Wilcoxon rank sum test: $W = 11{,}844$, $P = 0.009$). For each session, statistically significant differences in size existed as well between the colour morphs (see Supplementary Material Fig. S7). Shell length was

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Table 2. Main results of the tethering experiment.

Session	Colour morph	*n*	Shell length	Outcome				
				Alive	Preyed	Chipped	Empty shell	Lost (all)
Session 1	*Lineata* (autoch.)	33	7.15 ± 0.57	23	1	2	1	6
	Lineata (alloch.)	33	7.28 ± 0.62	24	2	3	1	3
	Nigra	33	7.64 ± 0.50	20	1	3	1	8
	Lutea	33	7.58 ± 0.68	16	6	6	1	4
	Total sample (session 1)	132	7.41 ± 0.62	83	10	14	4	21
Session 2	*Lineata* (autoch.)	38	6.81 ± 0.59	32	2	1	1	2
	Lineata (alloch.)	38	6.96 ± 0.56	31	5	1	1	0
	Nigra	38	7.60 ± 0.61	34	1	0	2	1
	Lutea	38	7.41 ± 0.45	30	0	3	2	3
	Total sample (session 2)	152	7.19 ± 0.64	127	8	5	6	6
Sessions 1 and 2 (whole experiment)	*Lineata* (autoch.)	71	6.97 ± 0.60	55	3	3	2	8
	Lineata (alloch.)	71	7.11 ± 0.60	55	7	4	2	3
	Nigra	71	7.62 ± 0.56	54	2	3	3	9
	Lutea	71	7.49 ± 0.58	46	6	9	3	7
	Total sample (sessions 1 and 2)	284	7.30 ± 0.64	210	18	19	10	27

The table shows the sample used (*n*), shell length (mean ± SD in mm) and main outcomes of the colour morphs used in each experimental session. '*Lineata* (autoch.)' is the resident snail from Cabo Silleiro and '*lineata* (alloch.)' is the transplanted *lineata* from Aguncheiro. The outcome 'alive' was obtained by summing the 'alive intact' and 'recovered alive intact' outcomes. 'Preyed' resulted from adding 'preyed alive' and 'preyed dead' (both indicative of crab predation attempts). 'Lost (all)' groups those outcomes in which the snail was lost or not recovered at the end of the experiment. Adding the results of 'preyed' and 'chipped' a new category 'preyed + chipped' was created for the logistic regression analyses (see the main text for further details)

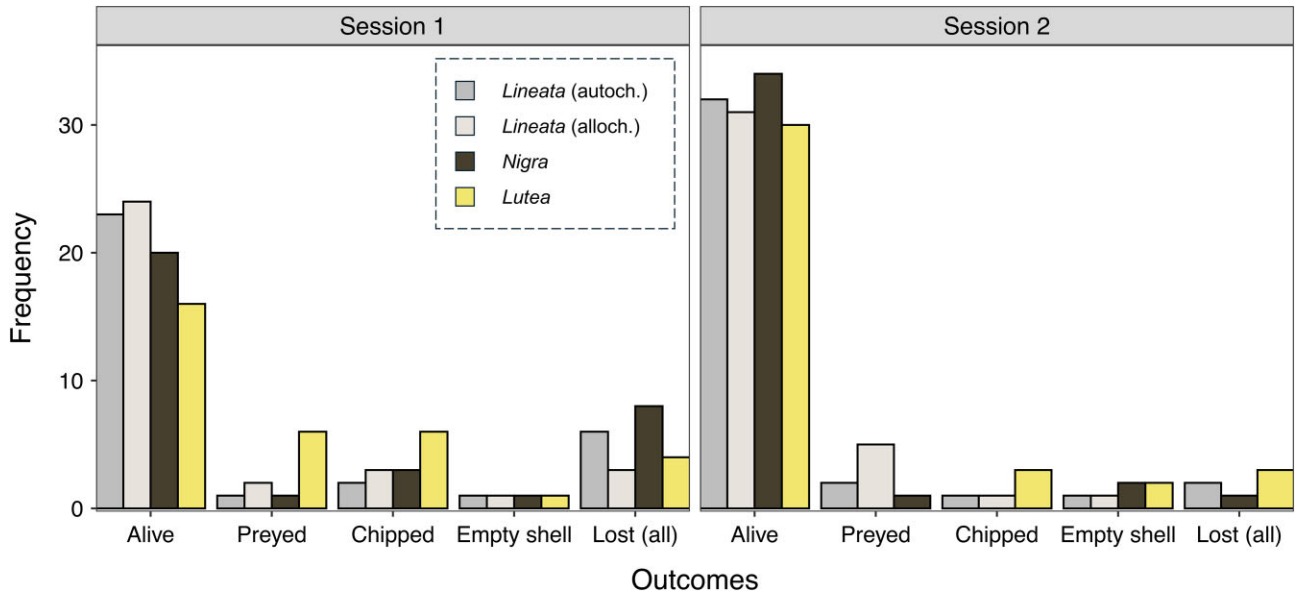


Figure 4. Histograms of main outcomes per colour morph for each experimental session.

always higher in the *lutea* and *nigra* snails than in either autochthonous or allochthonous *lineata*.

Since there were differences in the shell length of the snails between sessions, logistic regressions were used to explore the predictive power of this (i.e. 'shell length') and other potentially explanatory variables (particularly the colour of the morphs, but also 'session') on the probability of the different outcomes (see Table 3 and Supplementary Material Tables S2–S4). The resulting models confirmed 'session' and 'shell length' as explanatory variables of the outcome 'chipped' ($P < 0.05$; Supplementary Material Table S3), but not of 'preyed' or 'empty shell' (Supplementary Material Tables S2 and S4). In the 'chipped' model, 'transect position: lower' (whether the transect was located in the lower zone of the intertidal ecosystem) yielded a near-significant result ($P = 0.086$). 'Shell length' returned a positive relationship to the log odds of the probability of 'chipped', with an odds ratio of 2.29 (95% confidence interval = 1.04–5.23), while 'session 2' (reference category: 'session 1') and 'transect position: lower' (reference category: 'upper') resulted in a negative relationship (the latter, however, with an odds ratio 95% confidence interval including 1). These results can be interpreted as larger snails, snails of session 1 and perhaps snails of lower transects having a higher

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Table 3. Results of the logistic regression fitted to the composite 'preyed + chipped' outcome of the experiment.

Outcome: preyed + chipped							
Predictor	Level	Estimate (β)	SE	*Z*	*P*	Odds ratio	95% CI (odds ratio)
Intercept	–	−3.381	2.232	−1.52	0.1298	–	–
Shell length	–	0.237	0.306	0.77	0.4392	1.267	0.692–2.313
Session (reference: session 1)	**Session 2**	**−0.948**	**0.387**	**−2.45**	**0.0143**	**0.387**	**0.177–0.817**
Transect position (reference: lower)	Upper	−0.233	0.388	−0.60	0.5473	0.792	0.362–1.675
Colour morph (reference: *lineata* Cabo	*Lineata* Aguncheiro	0.576	0.552	1.04	0.2965	1.778	0.618–5.565
Silleiro—autoch.)	(alloch.)						
	Nigra	−0.290	0.666	−0.44	0.6629	0.748	0.194–2.781
	Lutea	0.973	0.549	1.77	0.0765	2.646	0.935–8.297
Model fit statistics							
AIC	209.4						
Null deviance	211.8						
Residual deviance	195.4						
Number of Fisher scoring iterations	5						

The model used 'shell length', 'session' (with 'session 1' as reference level), 'transect position' (with 'upper' as reference level) and 'colour morph' (with '*lineata* from Cabo Silleiro—autochthonous' as reference level) as predictors. The table provides the coefficient estimates (β), standard errors (SE) and *Z*- and *P*-values for each predictor or level, in addition to their corresponding odds ratios (e^{β}) and 95% confidence intervals (CI). The lower table displays the model fit statistics, which include the Akaike information criterion (AIC), the null and residual deviance, and the number of Fisher scoring iterations. Statistically significant results are presented in bold font. The 'preyed + chipped' outcome was created by granting the snails with 'preyed' or 'chipped' outcomes a value of '1', and '0' to those with the remaining outcomes. In all models, 'lost (all)' cases were considered NAs to avoid the 'missing as predation' bias.

probability of being 'chipped'. The contribution of 'shell length' may be related to the fact that the snails of session 1 were generally larger.

None of the models except that of 'preyed + chipped' conceded the colour morph of the snail a significant predictive role (Table 3). In that model, the colour morph *lutea* stood close to statistical significance, with a positive slope ($\beta = 0.97$, $P = 0.077$; reference level: *Lineata* autochthonous). The odds ratio suggests that having a shell of this colour increases the odds of being preyed on or having a small break in the shell by 2.65 times, but the corresponding 95% confidence interval includes 1 (0.94–8.30), so this value should be interpreted cautiously. 'Session 2' also yielded a significant result ($P = 0.014$) in the 'preyed + chipped' model (using 'session 1' as the reference category), with a negative estimate value and an odds ratio below 1 (95% confidence interval = 0.18–0.82), meaning that having participated in session 1 increases the probability of this predation outcome. When 'preyed + chipped' regression models were run for each session separately (Supplementary Material Table S5), in session 1 the colour morph *lutea* also yielded a significant result ($P = 0.04$; odds ratio = 4.56, 95% confidence interval = 1.17–22.93), but not in session 2 ($P = 0.82$; odds ratio = 1.23, 95% confidence interval = 0.19–8.12). While separate analyses of 'preyed' and 'chipped' counts by colour morph in session 1 did not yield statistically significant results (Fisher's exact test: 'preyed', $P = 0.136$; 'chipped', $P = 0.503$), the combined 'preyed + chipped' outcome did (Fisher's exact test: $P = 0.035$), with a higher count of *lutea* in the contingency table. When the experimental sessions were pooled, no differences were found between the different colour morphs in the frequencies of outcomes (Fisher's exact or chi-square tests, $P > 0.10$), except again in the additive category of 'preyed + chipped', which showed a near statistically significant result ($\chi^2 = 7.65$, $df = 3$, $P = 0.054$).

DISCUSSION

Ever since Francis B. Sumner's classic research on the coat colouration of *Peromyscus* mice (Sumner, 1926), colour clines have traditionally been investigated not only because colouration is a highly conspicuous and easily scorable trait but also because it poses interesting questions about the maintenance of colour variation (McLean & Stuart-Fox, 2014). Several instances of colour clines putatively maintained by selection have been described in molluscs and other animal groups (Antoniazza *et al.*, 2010; Brakefield & de Jong, 2011; Takahashi *et al.*, 2011). Similarly, the study of intrapopulation colour polymorphism has enhanced our understanding of adaptation, divergence and, ultimately, speciation (Comeault *et al.*, 2015; Svensson, 2017; O'Connor *et al.*, 2021). Various methods have been devised to study colour adaptation. Among these, frequentist-observational methods are typically the most used, with genomic approaches being added in recent decades (Takahashi, 2015; Cuthill *et al.*, 2017; San-Jose & Roulin, 2017).

The frequentist-observational procedure has yielded valuable insights into the adaptive role of colouration, substantially increasing our knowledge of this phenomenon (Gefaell *et al.*, 2023). However, more direct evidence is often needed to understand the precise ecological and evolutionary forces underlying variation in colouration, whether clinal or intrapopulation. In such cases, field experiments offer a powerful alternative. In field experiments, one or more potentially relevant causal factors are deliberately manipulated to assess and quantify their contribution to the phenomenon of interest. There are successful examples of field experiments in the evolutionary ecology of colouration (e.g. Ekendahl, 1998; Mullen & Hoekstra, 2008), but many of them involve several difficulties that can limit the validity of their results. First, many extraneous variables can rarely be controlled, potentially altering the results in unexpected ways. Second, these experiments rely on numerous untested assumptions about the natural history of the organisms involved. If any of these assumptions are not met, the results might be compromised. Even if the organisms taking part in the experiment have been thoroughly studied in the past, many relevant aspects of their behaviour in the wild are unknown, which can ultimately jeopardize field experiments. Finally, the intensity of the studied factor (such as selection against a morph) is often unknown in advance, complicating decisions about the necessary sample size to detect its effect.

In the present study, a tethering experiment aimed at testing whether there was differential crab predation of *Littorina saxatilis* colour morphs by *Pachygrapsus marmoratus* was conducted as a means of shedding light on the maintenance mechanisms of a colour cline in the populations of *L. saxatilis* from the Rías Baixas. More

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specifically, this experiment sought to determine why the wave-exposed localities along the cline tend to be monomorphic for *lineata*. As a complementary study, a series of samplings in the source and experiment localities (Aguncheiro and Cabo Silleiro), as well as a series of measurements of key shell parameters of the colour morphs involved, were carried out. Several insights can be gained from this study.

It appears that, at the source site (Aguncheiro), not all morphs are equally attacked by the crabs, with *lutea*, despite being less common, having a higher predation propensity than *nigra* and *lineata*, as inferred from the frequency of scars. This effect, however, is not statistically significant. Of these morphs, *lutea* appears to have a slightly thicker shell. Although this result should be taken cautiously due to its near-significance, it is unexpected, as previous studies suggest that no differences in shell thickness exist between colour morphs from the same site (Raffaelli, 1979). Shell thickness has traditionally been related to predation attempts, either constituting a genetic adaptation or a plastic response against predation (Seeley, 1986; Trussell, 1996; Bourdeau, 2009). In fact, healed predation attempts have been linked to the thickening of the shell, which might in turn reduce vulnerability to future crab attacks (Teck *et al.*, 2023). As for crypsis, the comparison of the spectra of the snails and rocks from Aguncheiro suggests that, in that site, *lutea* and *lineata* might be more cryptic than *nigra*. Given the chromatic diversity of Aguncheiro (with rocks with yellowish and black-reddish spots), it could be that each morph is adapted to a different colour patch (*lutea* to yellow patches, *lineata* to black-reddish ones), with *nigra* perhaps settling more on the bottom of the rocks, where light strikes less intensely and therefore can be more cryptic. If so, then the polymorphic population of Aguncheiro could be maintained by selection for crypsis in a heterogeneous environment. However, this hypothesis has to be further investigated to evaluate its veracity.

Reflectance spectrometry provides some support for the relatively high conspicuousness of *lutea* (and to some degree *nigra*) in a representative wave-exposed site from the Rías Baixas, where monomorphic populations of *lineata* abound. Although the degree of crypsis of the morphs should ideally be evaluated using visual models of the predator species (currently unavailable for *P. marmoratus*), the information provided by the spectra can be taken as provisional evidence for the hypothesis that, at Cabo Silleiro, the *lineata* morphs (especially the autochthonous one) are more cryptic than *lutea* and *nigra*. For instance, at *c.* 508 nm (the peak sensitivity of the main photoreceptor class of crabs), the spectra of *lineata* coincide with that of the rocks, something not seen in the remaining morphs (especially in *lutea*). With all due caution, this result provides frequentist-observational evidence for selection for crypsis as a mechanism contributing to the *lineata* monomorphic populations of *L. saxatilis* from the Rías Baixas. Reflectance spectrometry assessments of the degree of crypsis have rarely been used in *L. saxatilis* (see Gefaell *et al.*, 2023, and references therein, as well as the pioneering study by Cruickshanks, 2006). However, they do provide a useful way of testing a commonly assumed hypothesis in selection for crypsis studies, namely, that the different morphs vary in their level of camouflage.

As for the tethering experiment, very low predation rates were observed, substantially limiting the statistical power of the study. Indeed, a power analysis (see Poldrack, 2023) for the predictor 'colour morph' in the 'preyed + chipped' logistic regression (with likelihood ratio test with 1,000 simulations: $\alpha = 0.05$) revealed that the study was underpowered, as the power to detect a significant effect was 60.2% (95% confidence interval = 57.09–63.25%). Several possible explanations exist for the low predation rates obtained, all related to the previously discussed difficulties inherent to field experiments: either the lack of control over extraneous variables or the reliance on untested assumptions about the natural history of the organisms involved. One possibility might relate to the size of the snails used in the experiment (between 7 and 7.5 mm in shell length), which could have been too large to be preyed upon by the crabs. However, as stated previously, using much smaller snails would likely have masked their original colour, thus invalidating the experiment. Another possibility is related to our limited knowledge of the ecology and natural history of the predator species, *P. marmoratus*. It cannot be ruled out that the crabs were affected by some unknown factor that deterred them from preying on the snails. Alternatively, their densities might not have been as high as assumed during the weeks of the experiment. Measuring predation by crabs on *Littorina* snails, either in the laboratory or the field, is a demanding task that has faced many challenges in the past (Tucker, 1988; Ekendahl, 1998; Cruickshanks, 2006). For instance, in a well-designed laboratory experiment aimed at testing visual selection on *L. mariae* (now *L. fabalis*) colour morphs by the European green crab (*Carcinus maenas*), Tucker (1988, chapter 4) found no evidence of differential selection. However, the predation rates obtained were also very low, thus compromising the validity of the results.

Despite the low predation rates in our tethering experiment, we found some modest evidence of a selective disadvantage for the presumably less cryptic morph, *lutea*, against predation by *P. marmoratus*. In the first session, this morph was significantly more likely to be predated if both chips ('chipped' outcome) and more substantial breaks are counted as predation. This requires consideration of these factors ('chipped' and 'session') separately. The 'chipped' outcome has been previously interpreted as at least partly due to the snails being bashed against the rocks by the waves (Reimchen, 1982; Boulding *et al.*, 2017). In our case, this would predict more 'chipped' instances in session 1 (where there was bad weather during the first days) and in lower transects (as the snails tethered there would suffer more wave action). Evidence backs the first prediction (see the 'chipped' logistic regression, Supplementary Material Table S3), but the second one finds, at best, partial support, as differences in the 'chipped' outcome between transect locations are close to significance in both the regression models ($P = 0.086$) and a chi-square test ($\chi^2 = 3.06$, $df = 1$, $P = 0.080$). Alternatively, 'chipped' can be considered a predation attempt, most likely by a small crab. Favouring this interpretation is the fact that the outcome 'preyed', which under no circumstances can be taken as the result of bashing against the rocks, goes in the same direction as 'chipped' in the *lutea* morph from session 1 (see Fig. 4). This does not happen in the other morphs. Furthermore, if *lutea* is really a little bit thicker than *lineata* (both autochthonous and allochthonous) and *nigra*, then it would be expected that the shell breaks caused by the crabs in *lutea* would be smaller than those of the remaining morphs (thus contributing to a relatively higher frequency of 'chipped' in this morph). As for the differences among sessions, these may be due to duration. Session 1 lasted 11 days due to bad weather while session 2 lasted only 4 days. This would have given crabs in session 1 more time to prey upon the snails, as is seen in the comparison between the sum of the 'preyed' and 'chipped' outcome counts of both sessions (higher in session 1). However, if it were only a matter of time, we would expect to see the same trend in both sessions, which we did not, as no *lutea* individuals were preyed upon in session 2. So while it would be reasonable to assume that 'chipped' constitutes a predation attempt, and that predation in session 1 was higher because of the number of days it lasted, several uncertainties remain. It cannot be ruled out that 'chipped' includes both predation attempts by small crabs and breaks caused by wave action.

To summarize, no clear evidence of selection for crypsis was found in the polymorphic source population studied (Aguncheiro), as the asymmetrical distribution of shell scars between morphs is not statistically significant. Reflectance spectrometry has shown that, in the monomorphic *lineata* population of Cabo Silleiro, this morph is more cryptic than the allochthonous morphs, especially *lutea*. However, the tethering experiment conducted there provides only partial support for selection for crypsis, as evidence of predation against *lutea* was found only in one session of the experiment. In any case, due to low predation rates, the statistical power of the present study was rather low.

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In the face of the previous results, whether *P. marmoratus* plays a role in maintaining the monomorphic populations of *lineata* seen in the wave-exposed localities from the Rías Baixas by pruning the less cryptic morphs is currently unknown. Taken together, the evidence provided here should be viewed as suggestive yet inconclusive, calling for a future experiment aimed at definitely settling this question. Based on the data from the present study, a sample of 350 snails would suffice to confidently assess our hypothesis (power for predictor 'colour morph' using the likelihood ratio test, 1,000 simulations and an α level of 0.05 = 98%, with a 95% confidence interval of 96.93–98.77 and the simulated sample increased using sampling with replacement). Additionally, more than 4 days of experiment would be necessary to obtain enough predation instances to properly evaluate the hypothesis. In this context, using a larger sample size and longer experiment duration could help determine whether the monomorphic *lineata* populations in wave-exposed localities result from differential elimination of the more conspicuous morphs. These considerations should therefore guide the design of future tethering experiments of this organism.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Journal of Molluscan Studies* online.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

DATA AVAILABILITY

All data and software codes are openly available at https://github.com/juangefaell/TetheringPredation.git.

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